Resins: Resins are polymer made by repeatedly linking discrete molecules (monomers) together to form chains or networks.

~~chemical~~ properties of resins:-

① On heating, they soften and finally melt.

② Insoluble in water, petroleum spirit but dissolve in alcohol, chloroform & ether.

③ chemically resins are complex mixtures of resin acids, resin alcohols, resin phenols.

④ All resins are heavier than water, they are usually amorphous, hard and brittle solids.

⑤ Resins when burnt produce smoky flame.

Based upon their origin:-

Resins are two types -

① Natural resins:- is a product derived from plants.
Ex: GUM

② Synthetic resins: is typically produced via chemical synthesis.
Ex: Thermoplastics, Thermoset, synthetic elastomer.

Based on Thermal Behaviour:-

① Thermoplastics

② Thermoset.

Q: What are natural & synthetic resins? Mention importance of synthetic resins? (2.75)

## Natural resins:-

Natural resins are typically fusible and flammable organic substances that are transparent and are yellowish to brown in colour.

** They are formed in plant secretions, and are soluble in various organic liquid but not in water

** Most natural resins are exuded from trees, especially pines and ~~fies~~ firs.

Example:- amber, balsam, canada balsam, Dragon's blood from the dragon trees, elemi.

## Synthetic resins:-

Synthetic resin is typically produced via chemical synthesis. It's an industrially produced resins, typically viscous substances that convert into rigid polymers by the process of curing.

** The most commonly used type of synthetic resin is epoxy resin.

** They are used as thermoset polymer.

** They are not biodegradable.

(iii) Coordinate bonding:-

This types of bonding occurs also by the sharing of an electron pair but the shared electrons come from one atom called the donor.

The coordinate or semi-polar bond has properties between those of the ionic and covalent bonds.

Example:

Addition compounds of $BF_3$.




The coordinate bonds are not found in polymer.

* Rarely found in polymer. (Then lecture error)

(iv) Metallic bonding:-

In metallic bond, the no. of valance electron is far too small to provide complete shells for all the atoms.

This bond involves the positively charged atoms embedded in a permeating gas of electrons free to move about a high electrical conductivity.

Actually These types of bondings are not found in polymer.

Masud Azha
chem-44

Different types of bonding exists in polymers.

(i) Ionic (ii) covalent (iii) Coordinate (iv) Metallic.

## (i) Ionic bonding:-

Ionic bonds leads to the build up of large scale crystals.

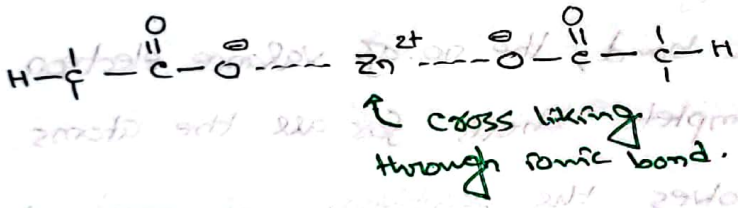
Ionic bonds are formed by donation of an electron by an atom to the This results in electrostatic charge on the atoms This electrostatic attraction bet positive & negative ion is called ionic bond.

$$Na \longrightarrow Na^+ + e^-$$

$$Cl + e \longrightarrow Cl^-$$

$$Na^+ + Cl^- \longrightarrow NaCl.$$

* Ionic bond are not usually found in polymer. except in the use of divalent ions to provide cross-links bet carbonyl groups in natural Rubber.




## (ii) Covalent bonding:-

This type of bonding occurs by sharing of an electron pairs by two atoms, resulting in stable electronic configuration The covalent bond is very frequent in polymer.



The covalent bond is predominant bond in polyme

Based on thermal processing behavior / ~~the~~ Heat stability :-

| Thermoplastic polymers | Thermosetting Polymer |
|---|---|
| ① These are the polymers which soften on heating & stiffen on cooling in order to process into desired form. | ① These are the polymers which on heating undergo chemical change & convert themselves into an ~~inso~~ infusible & insoluble mass. |
| ② There is no cross-linking between the chains. | ② There is cross-linking between the chains forming three dimensional network. |
| ③ By the application of heat & Pressure waste thermoplastic can be 'recovered' | ③ can not be thermally processed. |
| ④ can be recycled & reshaped | ④ cannot be recycled & reshaped. |
| ⑤ This is lower in molecular weight. | ⑤ This is high in molecular weight. |
| ⑥ It is synthesized by addition polymerization. | ⑥ It is synthesized by condensation polymerization. |
| ⑦ This has a low melting point, low tensile strength, rigidity, durability, brittleness. | ⑦ This has a high melting point, tensile strength, rigidity, durability, brittleness. |
| ⑧ Example: Polyethylene, pvc, nylon, polystyrene etc. | ⑧ Example: * Urea-formaldehyde, * melamine formaldehyde. |

...at are the organizational parameters of molecular ...ilding blocks for molecular materials in the core of organic and inorganic chemistry? (2)

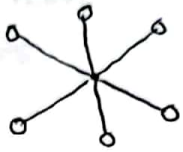
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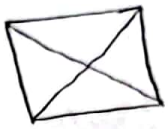
(3)

## Inorganic

* Types of order
* Strength & anisotropy of interaction bet' subunits.
* Symmetry of packing.
* Intermolecular vibration

## molecular building blocks

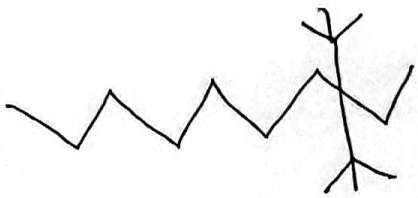




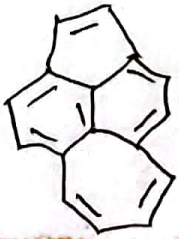
## organic

* Symmetry of molecules.
* Polarizability of a molecules.
* Degree of flexibility of submolecular component.
* Hydrophilic & hydrophobic properties.









## Supramolecule

② supramolecules are large structures that aren't necessarily covalently bonded together.

② The equivalent of low molecular weight organic molecules.

③ It holds together by non-covalent bond.

④ Host-guest chemistry

## macromolecule

① Macromolecules are covalently bonded monomers, like starch being made of multiple glucose subunits.

② It composed by about > 1000 atoms.

③ It holds together by covalent bonds.

④ supramolecular self-assembly